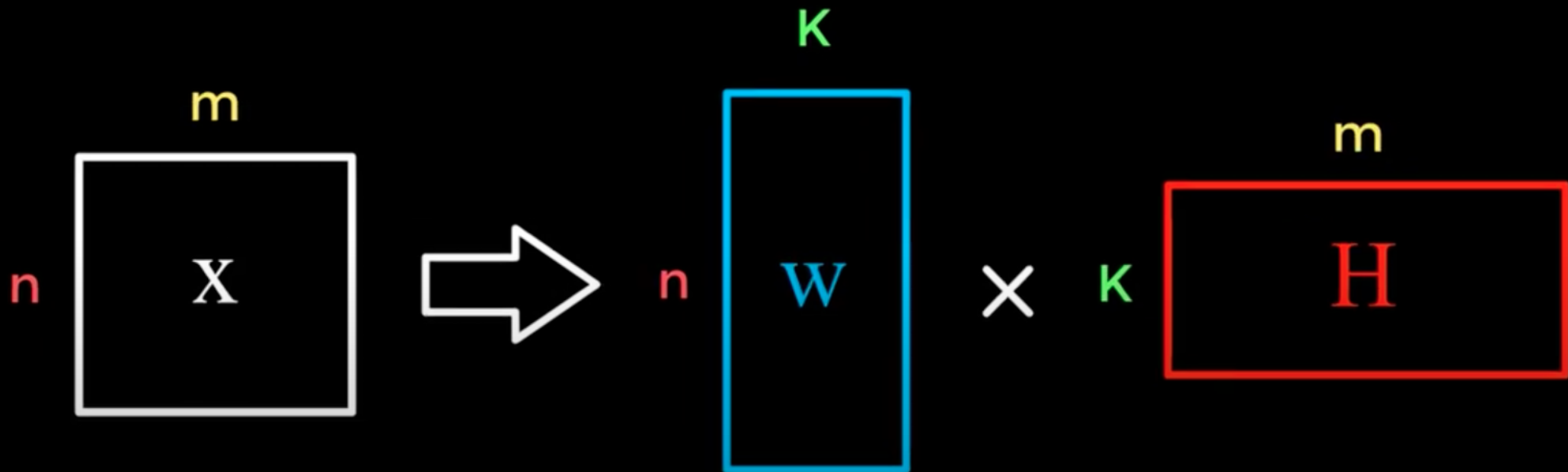
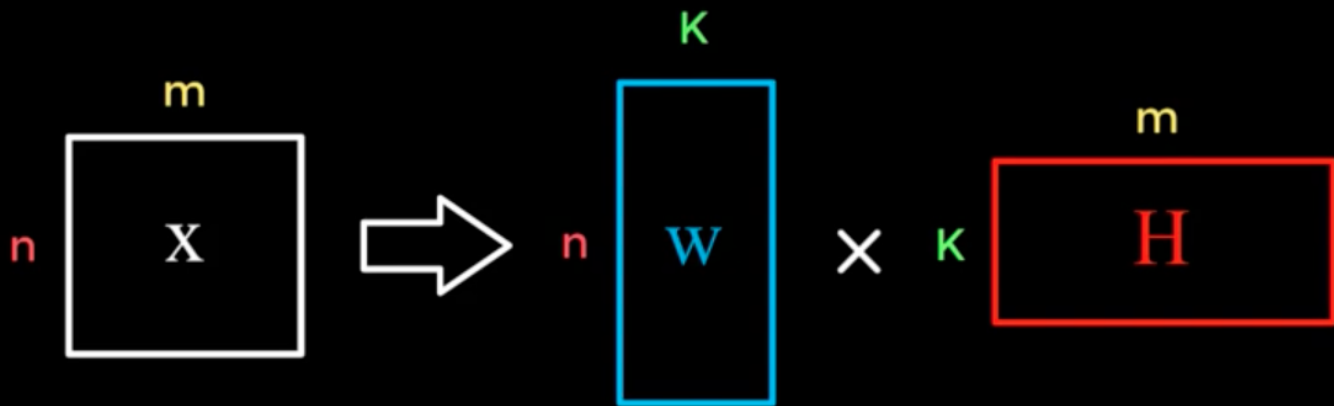


What is **NMF**?

Non-Negative Matrix Factorization





$$\text{minimize } ||X - W.H||^2, \text{ w.t. } W, H, \text{ s.t. } W, H > 0$$

NP-HARD PROBLEM!

Convex in W or H but not in Both!

$$\text{minimize } ||X - W.H||^2, \text{ w.t. } W, H, \text{ s.t. } W, H > 0$$

Consider the scalar case, where : $m=n=1$

$$\min(x - w.h)^2 = \min(x^2 - 2xwh + w^2h^2)$$

the gradient and hessian of $\phi_x(w, h) = x^2 - 2xwh + w^2h^2$ is :

$$\nabla \phi_x(w, h) = \begin{bmatrix} 2hw^2 - 2xw \\ 2h^2w - 2xh \end{bmatrix}$$

$$\nabla^2 \phi_x(w, h) = \begin{bmatrix} 2w^2 & 4hw - 2x \\ 4hw - 2x & 2h^2 \end{bmatrix}$$

**Not always
positive semi-definite !**

$$\text{minimize } ||X - W.H||^2, \text{ w.t. } W, H, \text{ s.t. } W, H > 0$$

Cannot be solved Analytically !

$$X \approx WH$$

A good local minimum can be satisfying for us.

Multiplicative Update

It is based on Gradient Descent with Multiplicative Update rules :

$$H \leftarrow H \odot \frac{W^T X}{W^T W H}$$

$$W \leftarrow W \odot \frac{W^T X}{W H H^T}$$

Multiplicative Update

It is based on Gradient Descent with Multiplicative Update rules :

$$H \leftarrow H - \eta \odot [W^T W H - W^T X] \quad \eta \rightarrow \frac{H}{W^T W H}$$

$$W \leftarrow W - \zeta \odot [W H H^T - X H^T] \quad \zeta \rightarrow \frac{W}{W H H^T}$$



$$\boxed{\begin{aligned} H &\leftarrow H \odot \frac{W^T X}{W^T W H} \\ W &\leftarrow W \odot \frac{W^T X}{W H H^T} \end{aligned}}$$